

INT-332 Practical-5

You are required to create a workflow configuration file(,GitHub/workflows/ci.yml) that performs the following tasks:

- 1. checks out hte source code from the repository.**
- 2. sets up the appropriate java development kit (jdk) version required for the project.**
- 3. installs all necessary project dependencies using maven.**
- 4. compiles the application and executs unit tests using appeopriate maven command.**
- 5.Archieve the generated build artifacts (such as JAR or WAR files) for future use.**

1. Create this workflow file in your project:

➤ `.github/workflows/ci.yml`

2.Add the following configuration:

```
name: Java Maven CI

on:
  push:
    branches:
      - main
  pull_request:
    branches:
      - main

jobs:
  build:

    runs-on: ubuntu-latest

    steps:
      # 1. Checkout source code
      - name: Checkout Repository
        uses: actions/checkout@v4

      # 2. Setup JDK
      - name: Set up JDK 17
        uses: actions/setup-java@v4
```

```

with:
  java-version: '17'
  distribution: 'temurin'

# 3. Cache Maven dependencies
- name: Cache Maven packages
  uses: actions/cache@v4
  with:
    path: ~/.m2
    key: ${{ runner.os }}-m2-${{ hashFiles('**/pom.xml') }}
    restore-keys: |
      ${{ runner.os }}-m2

# 4. Install dependencies, compile and run tests
- name: Build and Test with Maven
  run: mvn clean install

# 5. Archive generated JAR/WAR artifacts
- name: Upload Build Artifacts
  uses: actions/upload-artifact@v4
  with:
    name: java-build-artifacts
    path: |
      target/*.jar
      target/*.war

```

3. Checkout Repository

➤ `uses: actions/checkout@v4`

4. Setup JDK

➤ `uses: actions/setup-java@v4`

5. Cache Maven Dependencies

➤ `uses: actions/cache@v4`

6. Build and Test Application

➤ `run: mvn clean install`

7. Upload Build Artifacts

➤ `uses: actions/upload-artifact@v4`

The screenshot shows the GitHub Actions interface for the 'CI-CD Pipeline' workflow in the repository 'rajyadav23/docker-workflow'. The workflow file is 'ci-cd.yml'. The left sidebar shows the 'Actions' tab selected, with a list of workflow runs. The main area displays a table of 4 workflow runs, all on the 'main' branch. The runs are: 'fix backend build' (Commit 062b6f0, pushed by rajyadav23, Apr 9, 12:18 PM GMT+5:30, 12s), 'Fix Dockerfile name' (Commit 5576907, pushed by rajyadav23, Apr 8, 12:49 PM GMT+5:30, 18s), 'Fix CI docker compose issue' (Commit 75b25ca, pushed by rajyadav23, Apr 8, 12:44 PM GMT+5:30, 7s), and 'Initial CI/CD Docker project' (Commit e1d1111, pushed by rajyadav23, Apr 8, 12:34 PM GMT+5:30, 20s). All runs are marked with a red 'x' icon, indicating they failed.

Workflow Run	Commit	Pushed by	Branch	Status	Time
fix backend build	062b6f0	rajyadav23	main	Failed	12s
Fix Dockerfile name	5576907	rajyadav23	main	Failed	18s
Fix CI docker compose issue	75b25ca	rajyadav23	main	Failed	7s
Initial CI/CD Docker project	e1d1111	rajyadav23	main	Failed	20s

The screenshot shows the GitHub Actions interface for the 'Java CI with Maven' workflow in the repository 'rajyadav23/java-ci-maven'. The workflow file is 'java-ci-maven.yml'. The left sidebar shows the 'Actions' tab selected, with a list of workflow runs. The main area displays a table of 2 workflow runs, both on the 'main' branch. The runs are: 'Update maven.yml' (Commit 41f9647, pushed by rajyadav23, Apr 10, 12:47 PM GMT+5:30, 21s) and 'Create maven.yml' (Commit adc26fe, pushed by rajyadav23, Apr 10, 12:33 PM GMT+5:30, 25s). The first run is marked with a green checkmark, indicating it succeeded, while the second run is marked with a red 'x' icon, indicating it failed.

Workflow Run	Commit	Pushed by	Branch	Status	Time
Update maven.yml	41f9647	rajyadav23	main	Succeeded	21s
Create maven.yml	adc26fe	rajyadav23	main	Failed	25s